

Brooke Sevchik

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EDUCATION

Duke University, Durham, NC, USA

Bachelor of Science, Neuroscience

May 2024

Bachelor of Arts, Spanish, Latin American & Latino/a Studies

GPA: 4.0/4.0, *summa cum laude*

Relevant Coursework: Functional Neuroanatomy, Inside the Disordered Brain, Contemporary Neuroscience Methods, Introduction to Cognitive Neuroscience, Cellular and Molecular Neurobiology, Introduction to Biostatistics, Introduction to Computer Science

Coursework in Madrid, Spain

Universidad Carlos III de Madrid

Spring 2023

Duke in Madrid

Spring 2023, Summer 2022

RESEARCH EXPERIENCE

University of Pennsylvania Perelman School of Medicine, Penn Lifespan Informatics & Neuroimaging Center

Data Analyst, August 2024 - Present

Advisors: Dr. Theodore Satterthwaite, Dr. J. Cobb Scott, Dr. S. Parker Singleton

Leading open data release project on executive functioning in adolescents with ADHD/psychosis. Curating longitudinal dataset consisting of structural MRI, functional MRI, diffusion, and ASL images organized in BIDS format, as well as clinical and cognitive data for over 130 participants.

Co-lead a meta-analytic initiative, SYPRES (SYnthesis of Psychedelic REsearch Studies; <https://sympres.io/>), utilizing living systematic reviews to analyze randomized clinical trials of psychedelics for psychopathology symptoms and create an interactive online dashboard (<https://www.metapsy.org/sympres/>) with results.

Study coordinator for collection of 3T MRI data for myelin imaging study: Neuroimaging of Brain Structure and Function Across the Lifespan. Recruit and interact with participants, obtain MRI scans with MRI physicists and technologists.

Coordinated study start-up administrative processes following University of Pennsylvania Center for Advanced Magnetic Resonance Imaging and Spectroscopy regulations, and completed Level 1 MRI training.

Contribute to ongoing informatics hackathons on software for processing neuroimaging data, including CuBIDS, XCP-D, ASLPrep, and BABS softwares.

Duke University, Egner Lab at the Center for Cognitive Neuroscience

Research Assistant, February 2021 - May 2024

Advisors: Dr. Tobias Egner, Dr. Raphael Geddert

Spearheaded independent experiment to investigate effect of anxiety and threat on participants' learning of cognitive maps by conducting literature review, overseeing IRB approval, budgeting, programming experiment in Javascript, HTML, and CSS, collecting participant data, analyzing data in RStudio, and creating figures to present in posters and publications.

Earned highest possible score for senior honors thesis on this work, consisting of written paper and oral defense.

Recruited and interacted with participants on Amazon's Mechanical Turk; processed participant data utilizing statistical analyses in Python, RStudio and JASP. Created graphs and tables using RStudio to visually display results of experiments for use in scientific publications. Designed and programmed task variants with graduate student mentor for experiments on cognitive control and task switching.

Discussed the trajectory of research trials and investigated ideas for future experiments through lab discussions and literature reviews. Shadowed fMRI data acquisition protocols.

Duke University, Cogan Lab at the Duke Comprehensive Epilepsy Center*Research Assistant*, August 2022 - May 2024

Advisor: Dr. Gregory Cogan

Implemented response coding protocol consisting of filtering, labeling, and processing original audio files from ECoG studies using MATLAB and Audacity. Instructed new lab member on response coding technique. Discussed experimental design and analyzed scientific papers on neurolinguistics. Presented current literature on speech production and language models to graduate students and professors.

Duke University, Duke Institute for Brain Sciences, Summer Neuroscience Program*Fellow*, June 2023 - July 2023

Conducted full-time research in current labs and worked closely with faculty as part of an intensive paid research program. Developed soft skills in research communication, presentation efficacy, and scientific poster design during professional development seminar; elaborated and reflected on research conducted in written assignments.

Duke University, Center for Cognitive Neuroscience, Cognitive Neuroscience Research Internship (CNRI)*Intern*, January 2021 - April 2021

Collaborated with team of fellow interns selected for the competitive internship to plan final research project design on neuromarketing. Contributed to graduate student research projects through programming, creating stimulus sets, and discussing current scientific literature using skills learned in weekly Python and programming workshop. Analyzed neuroscience articles in journal club and presented current papers.

PUBLICATIONS**First-author**

Brooke L. Sevchik, Golia Shafiei, Kristin Murtha, Juliette B.H. Brook, Lia Brodrick, Kahini Mehta, Sage Rush, Matt Cieslak, Steven L. Meisler, Taylor Salo, S. Parker Singleton, Tien T. Tong, Mrugank Salunke, Dani S. Bassett, Monica E. Calkins, Mark A. Elliott, Raquel E. Gur, Ruben C. Gur, Tyler M. Moore, Kosha Ruparel, Russell T. Shinohara, M. Dylan Tisdall, Daniel H. Wolf, David R. Roalf, Theodore D. Satterthwaite, *Transdiagnostic Executive Function: An Open, Fully-Processed, Longitudinal Data Resource*. (*in preparation*)

S. Parker Singleton*, **Brooke L. Sevchik***, Analiese Lahey, Pim Cuijpers, Mathias Harrer, Megan T. Jones, Sandeep M. Nayak, Eric C. Strain, Simon N. Vandeckar, Robert H. Dworkin, J. Cobb Scott & Theodore D. Satterthwaite, *Psilocybin treatment for symptoms of depression: a living systematic review, meta-analysis, and data resource*, medRxiv (2025). doi: 10.1101/2025.08.13.25333530 *co-first authors

Brooke Sevchik, Raphael Geddert, and Tobias Egner, *Influence of Threat and Anxiety on Cognitive Map Learning*, PsyArXiv (2024). doi:10.31234/osf.io/u3cx7

Co-author

S. Parker Singleton, **Brooke L. Sevchik**, Simon N. Vandeckar, Eric C. Strain, Sandeep M. Nayak, Robert H. Dworkin, J. Cobb Scott & Theodore D. Satterthwaite, *An initiative for living evidence synthesis in clinical psychedelic research*, Nature Mental Health **3**, 3-5 (2025). doi:10.1038/s44220-024-00373-4

Juliette B.H. Brook, Taylor Salo, Audrey C. Luo, Joëlle Bagautdinova, Sage Rush, Aaron F. Alexander-Bloch, Erica B. Baller, Monica E. Calkins, Matt Cieslak, Elena C. Cooper, John Detre, Mark A. Elliot, Damien A. Fair, Phoebe Freedman, Philip R. Gehrman, Ruben C. Gur, Raquel E. Gur, Arno Klein, Nina Laney, Timothy O. Laumann, Kahini Mehta, Kathleen Merikangas, Michael P. Milham, Jonathan A. Mitchell, Tyler M. Moore, Steven M. Nelson, Kosha Ruparel, **Brooke L. Sevchik**, Sheila Shanmugan, Haochang Shou, Manuel Taso, Lauren K. White, Daniel H. Wolf, M. Dylan Tisdall, David R. Roalf, Theodore D. Satterthwaite. *An open, fully-processed data resource for studying mood and sleep variability in the developing brain*, bioRxiv (2025). doi:10.1101/2025.05.01.651544

CONFERENCE PRESENTATIONS

Brooke L. Sevchik, Golia Shafiei, Kristin Murtha, Juliette B.H. Brook, Lia Brodrick, Kahini Mehta, Sage Rush, Matt Cieslak, Steven L. Meisler, Taylor Salo, S. Parker Singleton, Tien T. Tong, Mrugank Salunke, Dani S. Bassett, Monica E. Calkins, Mark A. Elliott, Raquel E. Gur, Ruben C. Gur, Tyler M. Moore, Kosha Ruparel, Russell T. Shinohara, M. Dylan Tisdall, Daniel H. Wolf, David R. Roalf, Theodore D. Satterthwaite, *Transdiagnostic Executive Function: An Open, Fully-Processed, Longitudinal Data Resource*. Flux Congress, Dublin, IE, Fall 2025 (*presented poster*).

Brooke L. Sevchik, S. Parker Singleton, Simon N. Vandekar, Eric C. Strain, Sandeep M. Nayak, Robert H. Dworkin, J. Cobb Scott & Theodore D. Satterthwaite, *SYPRES: SYNthesis of Psychedelic REsearch Studies*. Society of Biological Psychiatry Annual Meeting, Toronto, CA, Spring 2025. (*presented poster*).

Brooke L. Sevchik, S. Parker Singleton, Simon N. Vandekar, Eric C. Strain, Sandeep M. Nayak, Robert H. Dworkin, J. Cobb Scott & Theodore D. Satterthwaite, *Living Evidence Synthesis of Psilocybin for Depression*. Dwight L. Evans, M.D., Penn Psychiatry Research Day, Translational Research: Bench to Bedside & Back Again. Philadelphia, PA, USA, Spring 2025. (*presented poster*).

Brooke Sevchik, Raphael Geddert, and Tobias Egner. *Influence of Threat and Anxiety on Cognitive Map Learning*. Cognitive Neuroscience Society Meeting, Toronto, CA, Spring 2024. (*presented poster*).

Brooke Sevchik, Raphael Geddert, and Tobias Egner. *Influence of Threat and Anxiety on Cognitive Map Learning*. Undergraduate Neuroscience Graduation with Distinction Poster Session. Durham, NC, Spring 2024. (*presented poster*).

Brooke Sevchik, Raphael Geddert, and Tobias Egner, *Influence of Threat and Anxiety on Cognitive Map Learning*. Duke University Summer Research Showcase. Durham, NC, Summer 2024. (*presented poster*).

ADDITIONAL CONFERENCE ATTENDANCES AND SOCIETY MEMBERSHIPS

Cognitive Neuroscience Society Meeting, Boston, MA, USA, Spring 2025.

Flux Congress, Baltimore, MD, USA, Fall 2024.

LEADERSHIP EXPERIENCE

Philip R. Costanzo Teaching Fellowship

Teaching Fellow, Academic Year 2023 - 2024

Created lesson plans, discussed techniques to improve learning among students, collaborated with teaching team to create study resources, and taught two Psychology 101 discussion sections each week. Advised and mentored students on semester-long research projects in class and during office hours. Graded student projects and essays. Engaged in training on teaching ethics, strategies, and psychology instruction and received feedback during weekly teaching seminar.

Duke University, Neurocare Mental Health Advocacy Organization

Student Outreach Chair, Treasurer, September 2021 - January 2023

Organized, facilitated, and recruited speakers for panels on women and mental health and imposter syndrome. Led meetings to plan and implement outreach events to promote mental health on campus and work with administration to address student concerns.

Duke University, Global Education Student Committee

Vice President of Marketing, Study Away Ambassador, February 2023 - May 2024

Promoted study abroad and shared personal experiences with prospective students, created marketing materials for events, trained student representatives on talking points, and coordinated with Global Education Office to distribute promotional materials.

Duke University, Residence Hall Community Council*Vice President*, September 2020 - April 2021

Advocated for student interests and facilitated communication among leadership and first-year students. Served as a representative of Southgate and Gilbert-Addoms Residence Halls on Duke University East Campus Council to plan, organize, and run social events for residents.

COMMUNITY SERVICE**Bleeding Disorders Foundation of North Carolina***Volunteer*, January 2020 - Present

Translate informational documents and flyers from English to Spanish and proofread documents.

Broadened nonprofit's rebranding efforts. Facilitated communication in Spanish about Zoom events during the pandemic.

El Futuro Mental Health Clinic, Durham, NC*Volunteer*, February 2022 - April 2022

Spearheaded design of, planned, and contributed to construction of a bilingual sensory garden on the mental health center's property.

HONORS AND GRANTS

Richard L. Predmore Award in Spanish, Duke University Department of Romance Studies	Spring 2024
Duke Undergraduate Research Support Conference Grant	Spring 2024
Phi Beta Kappa Honor Society Membership, Duke University Beta Chapter	Spring 2023
Psychology & Neuroscience Small Grant, Duke University Department of Psychology & Neuroscience	September 2022
Dean's List with Distinction, Duke University	Fall 2021, Fall 2022, Spring 2022, Spring 2024
Mac Anderson Foreign Language Scholarship, Duke University Global Education Office	Summer 2022
National Merit Scholarship Finalist	2019

MENTORSHIP**Tour Guide**, Duke University (April 2021 - July 2023)**Trinity Ambassador**, Duke University, Department of Romance Studies (June 2022 - May 2024)**TECHNICAL SKILLS AND INTERESTS**

Programming skills: RStudio, Python, Bash command-line programming, Javascript, HTML, CSS, JASP, Atom, collaborative coding through Git/GitHub (@B-Sevchik)

Additional training: Meta-Analysis in R Workshop, Physalia Courses.

Computer skills: Notion, Covidence meta-analysis software, Powerpoint/Microsoft Word/Excel, Google Slides/Google Docs/Google Sheets, Canva, Gimp image manipulation

Research skills: Completion of CITI program courses on human research ethics, experience with Duke's SONA undergraduate experiment system and Amazon's Mechanical Turk (MTurk), experience in IRB submission and amendment process, knowledge in handling PHI, BIDS and BIDS Apps, trained in rating MRI images for quality assessment

Academic interests: Impact of early life adversity on executive functioning, neuroimaging, impact of development on cognition, cognitive control, translational research to improve psychiatric symptoms, women's brain health and influence of puberty on development of psychopathology

LANGUAGES

Spanish: Advanced proficiency in reading & writing, Intermediate High level score for speaking on the American Council on the Teaching of Foreign Languages OPIc; Coursework on Spanish culture and experience living abroad in Madrid during Summer 2022 & Spring 2023